

Results

June 28, 2026

1 Tables of Friedman, Bonferroni-Dunn, Holm, Hochberg and Hommel Tests

Table 1: Average Rankings of the algorithms

Algorithm	Ranking
R2EC1	7.0107
R2EC2	4.8326
R2EC3	4.5835
R2EC4	4.4943
R2EC5	5.4699
R2EC6	4.3556
R2EC7	5.0852
R2EC8	4.6606
R2EC9	4.5076

Friedman statistic considering reduction performance (distributed according to chi-square with 8 degrees of freedom: 3667.4056799946015.

P-value computed by Friedman Test: 0.0.

Iman and Davenport statistic considering reduction performance (distributed according to F-distribution with 8 and 39992 degrees of freedom: 504.59818070845347.

P-value computed by Iman and Davenport Test: 0.0.

Table 2: Holm / Hochberg Table for $\alpha = 0.05$

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm/Hochberg/Hommel
8	R_{E-C1}	48.47527208107095	0.0	0.00625
7	R_{E-C5}	20.3442415276021	5.221349781075919E-02	0.0071428571428571435
6	R_{E-C3}	13.320612598527221	1.7566019706653289E-40	0.008333333333333333
5	R_{E-C7}	8.70878866433234	3.0713930075061583E-18	0.01
4	R_{E-C2}	5.568512667969572	2.5692295305643935E-8	0.0125
3	R_{E-C8}	4.160865695181366	3.170435016399969E-5	0.01666666666666666
2	R_{E-C3}	2.7751276246932006	0.00551800493241867	0.025
1	R_{E-C4}	2.5323039575325756	0.011331571475170252	0.05

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Bonferroni-Dunn's procedure rejects those hypotheses that have a p-value ≤ 0.00625 .

Hochberg's procedure rejects those hypotheses that have a p-value ≤ 0.05 .

Hommel's procedure rejects all hypotheses.

Table 3: Holm / Hochberg Table for $\alpha = 0.10$

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm/Hochberg/Hommel
8	R_{E-C1}	48.47527208107095	0.0	0.0125
7	R_{E-C5}	20.3442415276021	5.221349781075919E-02	0.014285714285714287
6	R_{E-C3}	13.320612598527221	1.7566019706653289E-40	0.01666666666666666
5	R_{E-C7}	8.70878866433234	3.0713930075061583E-18	0.02
4	R_{E-C2}	5.568512667969572	2.5692295305643935E-8	0.025
3	R_{E-C8}	4.160865695181366	3.170435016399969E-5	0.03333333333333333
2	R_{E-C3}	2.7751276246932006	0.00551800493241867	0.05
1	R_{E-C4}	2.5323039575325756	0.011331571475170252	0.1

Bonferroni-Dunn's procedure rejects those hypotheses that have a p-value ≤ 0.0125 .

Hochberg's procedure rejects those hypotheses that have a p-value ≤ 0.1 .

Hommel's procedure rejects all hypotheses.

Table 4: Adjusted p -values

i	algorithm	unadjusted p	P_{Bonf}	P_{Holm}	P_{Hoch}	P_{Homm}
1	R2E C_1	0.0	0.0	0.0	0.0	0.0
2	R2E C_5	5.221349781075919E-92	4.177079824860735E-91	3.654944846753143E-91	3.654944846753143E-91	3.654944846753143E-91
3	R2E C_7	1.7566019706653289E-40	1.4052815765322631E-39	1.0539611823991972E-39	1.0539611823991972E-39	1.0539611823991972E-39
4	R2E C_2	3.0713930075061583E-18	2.4571144060049266E-17	1.535696503753079E-17	1.535696503753079E-17	1.535696503753079E-17
5	R2E C_8	2.5692295305643935E-8	2.0553836244515148E-7	1.0276918122257574E-7	1.0276918122257574E-7	1.0276918122257574E-7
6	R2E C_3	3.170435016399969E-5	2.5363480131199753E-4	9.511305049199907E-5	9.511305049199907E-5	9.511305049199907E-5
7	R2E C_9	0.00551800493241867	0.04414403945934936	0.01103600986483734	0.01103600986483734	0.01103600986483734
8	R2E C_4	0.011331571475170252	0.09065257180136202	0.011331571475170252	0.011331571475170252	0.011331571475170252

Table 5: Holm / Shaffer Table for $\alpha = 0.05$

i	algorithms	$z = (R_0 - R_i)/SE$	p	Holm	Shaffer
36	R2E C_1 vs. R2E C_6	48.47527208107095	0.0	0.001388888888888889	0.001388888888888889
35	R2E C_1 vs. R2E C_4	45.94296812353838	0.0	0.0014285714285714286	0.0017857142857142859
34	R2E C_1 vs. R2E C_9	45.70014445637775	0.0	0.0014705882352941176	0.0017857142857142859
33	R2E C_1 vs. R2E C_3	44.31440638588958	0.0	0.0015151515151515152	0.0017857142857142859
32	R2E C_1 vs. R2E C_8	42.906759413101376	0.0	0.0015625	0.0017857142857142859
31	R2E C_1 vs. R2E C_2	39.766483416737714	0.0	0.0016129032258064516	0.0017857142857142859
30	R2E C_1 vs. R2E C_7	35.15465948254373	9.866670717600595E-271	0.0016666666666666668	0.0017857142857142859
29	R2E C_1 vs. R2E C_5	28.13103055346885	4.089578394990942E-174	0.001724137931034483	0.0017857142857142859
28	R2E C_5 vs. R2E C_6	20.3442415276021	5.221349781075919E-92	0.0017857142857142859	0.0017857142857142859
27	R2E C_4 vs. R2E C_5	17.811937570069528	5.7098564791547304E-71	0.001851851851851852	0.002272727272727273
26	R2E C_5 vs. R2E C_9	17.5691139029089	4.247548916377442E-69	0.0019230769230769232	0.002272727272727273
25	R2E C_3 vs. R2E C_5	16.183375832420737	6.607038278379986E-59	0.002	0.002272727272727273
24	R2E C_5 vs. R2E C_8	14.77572885963253	2.1007229753337517E-49	0.0020833333333333333	0.002272727272727273
23	R2E C_6 vs. R2E C_7	13.320612598527221	1.7566019706653289E-40	0.002173913043478261	0.002272727272727273
22	R2E C_2 vs. R2E C_5	11.635452863268869	2.7214189230543527E-31	0.002272727272727273	0.002272727272727273
21	R2E C_4 vs. R2E C_7	10.788308640994645	3.909176247002632E-27	0.002380952380952381	0.002380952380952381
20	R2E C_7 vs. R2E C_9	10.54548497383402	5.329716722256572E-26	0.0025	0.002777777777777778
19	R2E C_3 vs. R2E C_7	9.159746903345855	5.201917688280371E-20	0.002631578947368421	0.002777777777777778
18	R2E C_2 vs. R2E C_6	8.708788664333234	3.0713930075061583E-18	0.002777777777777778	0.002777777777777778
17	R2E C_7 vs. R2E C_8	7.752099930557648	9.038521439587544E-15	0.0029411764705882353	0.003125
16	R2E C_5 vs. R2E C_7	7.023628929074881	2.1617847765519237E-12	0.003125	0.003125
15	R2E C_2 vs. R2E C_4	6.176484706800657	6.554455535215884E-10	0.0033333333333333335	0.0033333333333333335
14	R2E C_2 vs. R2E C_9	5.933661039640032	2.9625313726852146E-9	0.0035714285714285718	0.0038461538461538464
13	R2E C_6 vs. R2E C_8	5.568512667969572	2.5692295305643935E-8	0.0038461538461538464	0.0038461538461538464
12	R2E C_2 vs. R2E C_3	4.611823934193988	3.991510219648545E-6	0.004166666666666667	0.004166666666666667
11	R2E C_2 vs. R2E C_7	4.547922969151867	5.417795733120329E-6	0.004545454545454546	0.004545454545454546
10	R2E C_3 vs. R2E C_6	4.160865695181366	3.170435016399969E-5	0.005	0.005
9	R2E C_2 vs. R2E C_8	3.1402759963636604	0.0016878873464514503	0.005555555555555556	0.005555555555555556
8	R2E C_4 vs. R2E C_8	3.036208710436997	0.002395734682330964	0.00625	0.00625
7	R2E C_8 vs. R2E C_9	2.793385043276372	0.005215956453622641	0.0071428571428571435	0.0071428571428571435
6	R2E C_6 vs. R2E C_9	2.7751276246932006	0.00551800493241867	0.0083333333333333333	0.0083333333333333333
5	R2E C_4 vs. R2E C_6	2.5323039575325756	0.011331571475170252	0.01	0.01
4	R2E C_3 vs. R2E C_4	1.6285617376487902	0.10340582973121656	0.0125	0.0125
3	R2E C_3 vs. R2E C_8	1.407646972788207	0.15923563095906312	0.016666666666666666	0.016666666666666666
2	R2E C_3 vs. R2E C_9	1.385738070488165	0.1658268987664532	0.025	0.025
1	R2E C_4 vs. R2E C_9	0.24282366716062528	0.8081420004512727	0.05	0.05

Nemenyi's procedure rejects those hypotheses that have a p-value $\leq 0.001388888888888889$.
Holm's procedure rejects those hypotheses that have a p-value ≤ 0.01 .
Shaffer's procedure rejects those hypotheses that have a p-value $\leq 0.001388888888888889$.

Table 6: Holm / Shaffer Table for $\alpha = 0.10$

i	algorithms	$z = (R_0 - R_i) / SE$	p	Holm	Shaffer
36	R2E C1 vs. R2E C6	48.473272081070895	0.0	0.0027777777777777778	0.0027777777777777778
35	R2E C1 vs. R2E C4	45.94296812353838	0.0	0.002857142857142857	0.0035714285714285718
34	R2E C1 vs. R2E C9	45.70014445637775	0.0	0.0029411764705882353	0.0035714285714285718
33	R2E C1 vs. R2E C3	44.31440638588958	0.0	0.0030303030303030303	0.0035714285714285718
32	R2E C1 vs. R2E C8	42.906759413101376	0.0	0.003125	0.0035714285714285718
31	R2E C1 vs. R2E C2	39.766483416737714	0.0	0.0032258064516129032	0.0035714285714285718
30	R2E C1 vs. R2E C7	35.15465948254373	9.866670717600595E-271	0.0033333333333333335	0.0035714285714285718
29	R2E C1 vs. R2E C5	28.13103055346885	4.089578394909042E-174	0.0033448275862068966	0.0035714285714285718
28	R2E C5 vs. R2E C6	20.3442415276021	5.221349781075919E-92	0.0035714285714285718	0.0035714285714285718
27	R2E C4 vs. R2E C5	17.811937570069528	5.7098564791547304E-71	0.003703703703703704	0.004545454545454546
26	R2E C5 vs. R2E C9	17.5691139029089	4.247548916377442E-69	0.0038461538461538464	0.004545454545454546
25	R2E C3 vs. R2E C5	16.183375832420737	6.607038278379986E-59	0.004	0.004545454545454546
24	R2E C5 vs. R2E C8	14.77572885963253	2.1007229753337517E-49	0.004166666666666667	0.004545454545454546
23	R2E C6 vs. R2E C7	13.320612598527221	1.7566019706653289E-40	0.004347826086956522	0.004545454545454546
22	R2E C2 vs. R2E C7	11.63545263268869	2.7214189230543527E-31	0.004545454545454546	0.004545454545454546
21	R2E C4 vs. R2E C7	10.788308640994645	3.909176247002632E-27	0.004761904761904762	0.004761904761904762
20	R2E C7 vs. R2E C7	10.54548497383402	5.329716722256572E-26	0.005	0.005555555555555556
19	R2E C3 vs. R2E C7	9.159746903345855	3.0713930075061983E-18	0.005263157894736842	0.005555555555555556
18	R2E C2 vs. R2E C6	8.70878664333234	5.201917688280371E-20	0.005555555555555556	0.005555555555555556
17	R2E C7 vs. R2E C8	7.752099305576481	9.038521439987544E-15	0.0058823529411764705	0.00625
16	R2E C5 vs. R2E C7	7.023628929074881	2.1617847765519237E-12	0.00625	0.00625
15	R2E C2 vs. R2E C4	6.176484706800657	6.554455535215884E-10	0.006666666666666667	0.006666666666666667
14	R2E C2 vs. R2E C9	5.933661039640032	2.9625313726852146E-9	0.0071428571428571435	0.007692307692307693
13	R2E C6 vs. R2E C9	5.568512667969572	2.5692295305643935E-8	0.007692307692307693	0.007692307692307693
12	R2E C2 vs. R2E C7	4.611823934193988	3.991510219648545E-6	0.008333333333333333	0.008333333333333333
11	R2E C2 vs. R2E C7	4.547922969151867	5.417795733120329E-6	0.009090909090909092	0.009090909090909092
10	R2E C3 vs. R2E C6	4.160865695181366	3.170435016399969E-5	0.01	0.01
9	R2E C2 vs. R2E C8	3.1402759963636604	0.001687873464514503	0.011111111111111112	0.011111111111111112
8	R2E C4 vs. R2E C8	3.036208710436997	0.002395734682330964	0.0125	0.0125
7	R2E C8 vs. R2E C9	2.793385043276372	0.005215956453622641	0.0125	0.0125
6	R2E C6 vs. R2E C9	2.7751276246932006	0.00551800493241867	0.016666666666666666	0.016666666666666666
5	R2E C4 vs. R2E C6	2.532039575325756	0.011331571475170252	0.02	0.02
4	R2E C3 vs. R2E C4	1.6285617376487902	0.10340582973121656	0.025	0.025
3	R2E C3 vs. R2E C8	1.407646972788207	0.15923563095906312	0.03333333333333333	0.03333333333333333
2	R2E C3 vs. R2E C9	1.385738070488165	0.165826897664532	0.05	0.05
1	R2E C4 vs. R2E C9	0.24282366716062528	0.8081420004512727	0.1	0.1

Nemenyi's procedure rejects those hypotheses that have a p-value $\leq 0.0027777777777777778$.
Holm's procedure rejects those hypotheses that have a p-value ≤ 0.025 .
Shaffer's procedure rejects those hypotheses that have a p-value $\leq 0.0027777777777777778$.

Table 7: Adjusted p -values

i	hypothesis	unadjusted p	p_{Neme}	p_{Holm}	p_{Shaf}	p_{Berg}
1	R2E _{C1} vs. R2E _{C6}	.0000	.0000	.0000	.0000	.0000
2	R2E _{C1} vs. R2E _{C4}	.0000	.0000	.0000	.0000	.0000
3	R2E _{C1} vs. R2E _{C9}	.0000	.0000	.0000	.0000	.0000
4	R2E _{C1} vs. R2E _{C3}	.0000	.0000	.0000	.0000	.0000
5	R2E _{C1} vs. R2E _{C8}	.0000	.0000	.0000	.0000	.0000
6	R2E _{C1} vs. R2E _{C2}	.0000	.0000	.0000	.0000	.0000
7	R2E _{C1} vs. R2E _{C7}	.0000	.0000	.0000	.0000	.0000
8	R2E _{C1} vs. R2E _{C5}	.0000	.0000	.0000	.0000	.0000
9	R2E _{C5} vs. R2E _{C6}	.0000	.0000	.0000	.0000	.0000
10	R2E _{C4} vs. R2E _{C5}	.0000	.0000	.0000	.0000	.0000
11	R2E _{C5} vs. R2E _{C9}	.0000	.0000	.0000	.0000	.0000
12	R2E _{C3} vs. R2E _{C5}	.0000	.0000	.0000	.0000	.0000
13	R2E _{C5} vs. R2E _{C8}	.0000	.0000	.0000	.0000	.0000
14	R2E _{C6} vs. R2E _{C7}	.0000	.0000	.0000	.0000	.0000
15	R2E _{C2} vs. R2E _{C5}	.0000	.0000	.0000	.0000	.0000
16	R2E _{C4} vs. R2E _{C7}	.0000	.0000	.0000	.0000	.0000
17	R2E _{C7} vs. R2E _{C9}	.0000	.0000	.0000	.0000	.0000
18	R2E _{C3} vs. R2E _{C7}	.0000	.0000	.0000	.0000	.0000
19	R2E _{C2} vs. R2E _{C6}	.0000	.0000	.0000	.0000	.0000
20	R2E _{C7} vs. R2E _{C8}	.0000	.0000	.0000	.0000	.0000
21	R2E _{C5} vs. R2E _{C7}	.0000	.0000	.0000	.0000	.0000
22	R2E _{C2} vs. R2E _{C4}	.0000	.0000	.0000	.0000	.0000
23	R2E _{C2} vs. R2E _{C9}	.0000	.0000	.0000	.0000	.0000
24	R2E _{C6} vs. R2E _{C8}	.0000	.0000	.0000	.0000	.0000
25	R2E _{C2} vs. R2E _{C7}	.0000	.0001	.0000	.0000	.0000
26	R2E _{C2} vs. R2E _{C3}	.0000	.0002	.0001	.0001	.0000
27	R2E _{C3} vs. R2E _{C6}	.0000	.0011	.0003	.0003	.0000
28	R2E _{C2} vs. R2E _{C8}	.0017	.0608	.0152	.0152	.0000
29	R2E _{C4} vs. R2E _{C8}	.0024	.0862	.0192	.0192	.0000
30	R2E _{C8} vs. R2E _{C9}	.0052	.1878	.0365	.0365	.0000
31	R2E _{C6} vs. R2E _{C9}	.0055	.1986	.0365	.0365	.0000
32	R2E _{C4} vs. R2E _{C6}	.0113	.4079	.0567	.0567	.0000
33	R2E _{C3} vs. R2E _{C4}	.1034	3.7226	.4136	.4136	.0000
34	R2E _{C3} vs. R2E _{C8}	.1592	5.7325	.4777	.4777	.0000
35	R2E _{C3} vs. R2E _{C9}	.1658	5.9698	.4777	.4777	.0000
36	R2E _{C4} vs. R2E _{C9}	.8081	29.0931	.8081	.8081	.0000